

Sarah Tise

Ergonomic Design:

Computer Mouse Analysis



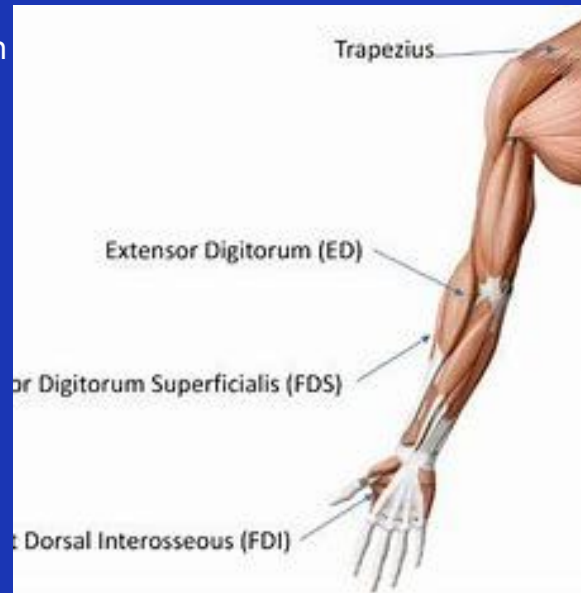
Product Focus:

Typical Engineer workspace has many tools, but this research will focus on the **computer mouse**.



Product Focus:

- **How does this tool come into contact with the human body?**
 - Wrist, hand, fingers
 - Many muscles in arm are used to operate a computer mouse, but this research will focus on the wrist area
 - Overall body posture does matter but this research will mostly focus on wrist posture



Research Hypothesis:

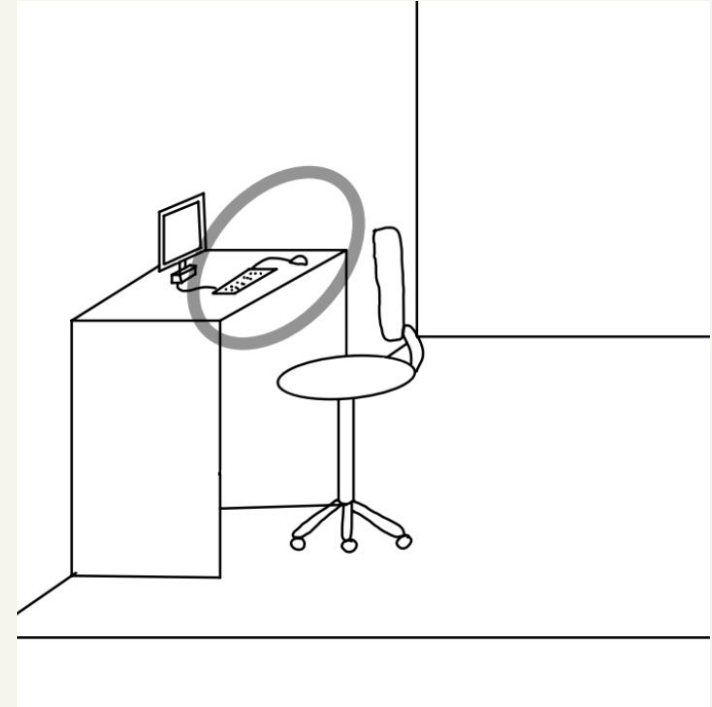
Office jobs such as engineering that often require a significant amount of time spent working on a computer put workers at a higher risk for developing certain musculoskeletal conditions. While current innovations such as the computer mouse make a world of a different when it comes to various computer heavy tasks that an engineer may have to complete during a typical work day, such as CAD drafting, more improvements which promote an ergonomic workspace while not hindering productivity and effectiveness can be made. Implementing ergonomic interventions in engineer offices such as adjustable desks, and chairs, or wrist supports, can significantly reduce the risk of developing conditions such as carpal tunnel syndrome.

Research Hypothesis Breakdown:

- **Research Hypothesis & how it complicates work activity:**
 - Implementing ergonomic interventions in engineer offices such as adjustable desks, and chairs, or wrist supports, can significantly reduce the risk of developing conditions such as carpal tunnel syndrome.
 - Computer mouse - puts wrist in uncomfortable position
 - Complications - computer mouse is key for CAD work so redesign can't be taken lightly
- **Research Hypothesis & Improvements to be made:**
 - Prioritize giving wrist more support
 - Wrist is parallel with tabletop, less tension in hand muscles & wrist joints

Workplace Description:

- The image shows what the workplace these research focuses on looks like.
 - Typically 1 worker(ie cubicles)
 - Recent college grads to older adults
 - Mostly men & some women
 - Typically 40 hours a week
 - Computer work - mostly done sitting down & with computer mouse
 - CAD
 - Calculations
 - Excel



Workplace & Wrist Posture:

- **Current Design Shortcomings:**
 - Allows wrist to bend at awkward/uncomfortable angles rather than stay parallel with tabletop
 - Puts too much pressure on nerves



Methodology:

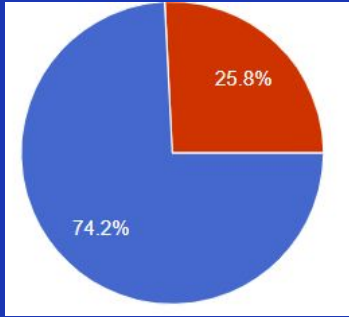
- **Review tasks for risk factors:**
 - Collected data from engineering students about discomfort experienced when spending time at computers
- **Control risk factors:**
 - Created prototype meant to support wrist while using a computer mouse
- **Making a working, ergonomic space:**
 - Conduct specific research on what components of engineer work space are absolutely necessary & how much ergonomic support these tools provide
- **Improving ergonomics via work station design principles:**
 - Applied ergonomic design principles:
 - Minimal contact stress, adjustability, design for diverse users, & neutral position in order to improve overall work station layout for CAD heavy work done by engineers

Student Engineer Survey:

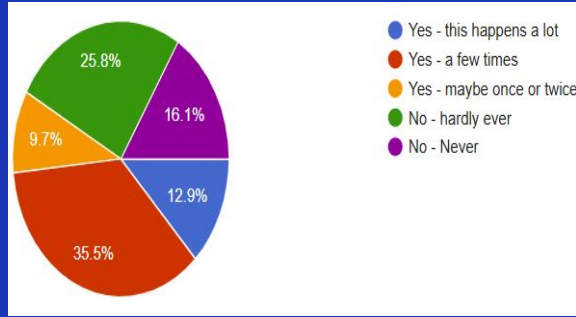
- **Purpose of survey:**
 - To get a better understanding of how relevant this topic is to engineers
 - To gather what the general consensus among engineers is on this topic
- **Who was surveyed?**
 - Engineering students at UIC
 - 31 random students of the COE at UIC
- **Why This group?**
 - Convenient access
 - Younger people are less likely to experience musculoskeletal pain so if survey found that this age group did have significant experience with said pain, this would be significant



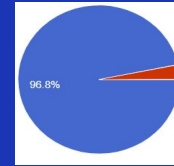
Collected Data: Engineer Survey



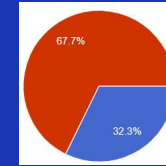
>70% of Engineering students at UIC use a computer mouse for at least 1 hour everyday



Significant amount of engineering students experience wrist pain after using a mouse for prolonged time



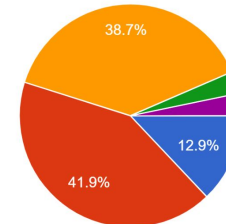
>96% of surveyed engineering students believe ergonomic workspace should be a top priority



>67 % Engineering Students believe UIC does not prioritize creating ergonomic work spaces for students

How ergonomic do you think computer mice are in general?

31 responses



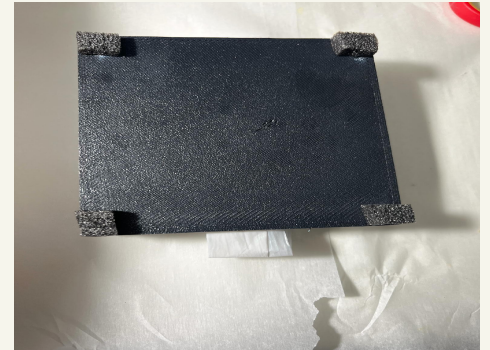
- 5 - very comfortable, no improvement needed
- 4 - mostly comfortable, some minor improvements could be beneficial
- 3 - somewhat comfortable, improvements suggested
- 2 - uncomfortable, improvements are necessary
- 1 - very uncomfortable, improvements should be an urgent priority

Development Improvement:

- **Redesign to reduce cost**
 - Affordable 3D printing
 - 3D printed prototype wrist support out of most affordable materials available
 - Used inexpensive foam, plastic, and super glue for other parts of tool
- **Redesign to improve productivity**
 - Easy motion of mouse
 - Ensured low friction between wrist support & table top so that tool would not lessen productivity & effectiveness of computer mouse
- **Redesign to improve quality**
 - PLA
 - PLA is the most affordable 3D printing material that was accessible & it is still a quality material
- **Redesign for better safety culture**
 - Foam glued on top to prevent irritation of wrist skin
 - Foam glued on bottom to prevent any scratching of table top as well

Development Improvement:

- The images here show the latest prototype as described on the previous slide



Development Improvement: Old Designs

Prototype 1: No PLA →



Prototype 2: No "bumpers" →



REBA/RULA Assessment:

#1

+

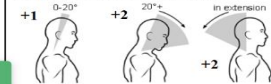
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REBA Employee Assessment Worksheet

based on Technical note: Rapid Entire Body Assessment (REBA), Hignett, McAtamney, Applied Ergonomics 31 (2000) 201-205

A. Neck, Trunk and Leg Analysis

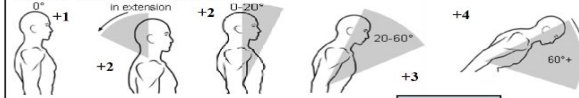
Step 1: Locate Neck Position



Step 1a: Adjust...
If neck is twisted: +1
If neck is side bending: +1

Enter text X
+2
Neck Score

Step 2: Locate Trunk Position



Step 2a: Adjust...
If trunk is twisted: +1
If trunk is side bending: +1

+2
Trunk Score

Step 3: Legs



Step 4: Look-up Posture Score in Table A
Using values from steps 1-3 above, locate score in Table A

Step 5: Add Force/Load Score

If load < 11 lbs : +0
If load 11 to 22 lbs : +1
If load > 22 lbs : +2
Adjust: If shock or rapid build up of force: add +1

Step 6: Score A, Find Row in Table C

Add values from steps 4 & 5 to obtain Score A.
Find Row in Table C:

Scoring:

1 = negligible risk
2 or 3 = low risk, change may be needed
4 to 7 = medium risk, further investigation, change soon

SCORES

Table A		Neck											
		1				2				3			
Trunk Posture Score	Legs	1	2	3	4	1	2	3	4	1	2	3	4
	1	1	2	3	4	1	2	3	4	3	3	5	6
	2	2	3	4	5	3	4	5	6	4	5	6	7
	3	2	4	5	6	4	5	6	7	5	6	7	8
	4	3	5	6	7	5	6	7	8	6	7	8	9

Table B		Lower Arm					
		1			2		
Upper Arm Score	Wrist	1	2	3	1	2	3
	1	1	2	2	1	2	3
	2	1	2	3	2	3	4
	3	3	4	5	4	5	5
	4	4	5	5	5	6	7
	5	6	7	8	7	8	8

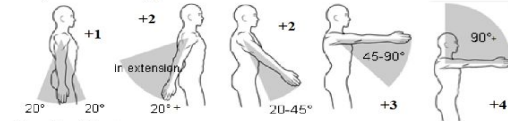
Table C		Score B, (table B value + coupling score)											
		1	2	3	4	5	6	7	8	9	10	11	12
Score A (score from table A + load/force score)	1	1	1	1	2	3	3	4	5	6	7	7	7
	2	1	2	2	3	4	4	5	6	6	7	7	8
	3	2	3	3	3	4	5	6	7	7	8	8	8
	4	3	4	4	4	5	6	7	8	8	9	9	9
	5	4	4	4	5	6	7	8	8	9	9	9	9
	6	6	6	6	7	8	8	9	9	10	10	10	10
	7	7	7	7	8	9	9	9	10	10	11	11	11
	8	8	8	8	9	10	10	10	10	11	11	11	11
	9	9	9	9	10	10	10	11	11	11	12	12	12
	10	10	10	10	11	11	11	11	12	12	12	12	12
	11	11	11	11	11	12	12	12	12	12	12	12	12
	12	12	12	12	12	12	12	12	12	12	12	12	12

3 + 3
Table C Score + Activity Score

6

B. Arm and Wrist Analysis

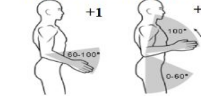
Step 7: Locate Upper Arm Position:



Step 7a: Adjust...
If shoulder is raised: +1
If upper arm is abducted: +1
If arm is supported or person is leaning: -1

+2
Upper Arm Score

Step 8: Locate Lower Arm Position:



+1
Lower Arm Score

Step 9: Locate Wrist Position:



+1
Wrist Score

Step 9a: Adjust...
If wrist is bent from midline or twisted: Add +1

Step 10: Look-up Posture Score in Table B

Using values from steps 7-9 above, locate score in Table B

1
Posture Score B

Step 11: Add Coupling Score

Well fitting Handle and mid rang power grip, **good**: +0
Acceptable but not ideal hand hold or coupling acceptable with another body part, **fair**: +1
Hand hold not acceptable but possible, **poor**: +2
No handles, awkward, unsafe with any body part, **Unacceptable**: +3

+
Coupling Score

Step 12: Score B, Find Column in Table C

Add values from steps 10 & 11 to obtain Score B. Find column in Table C and match with Score A in row from step 6 to obtain Table C Score.

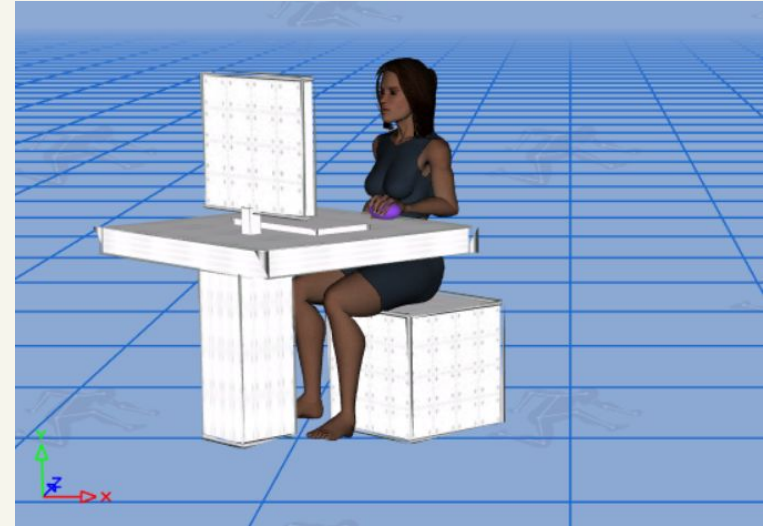
2
Score B

Step 13: Activity Score

+1 1 or more body parts are held for longer than 1 minute (static)
+1 Repeated small range actions (more than 4x per minute)
+1 Action causes rapid large range changes in postures or unstable base

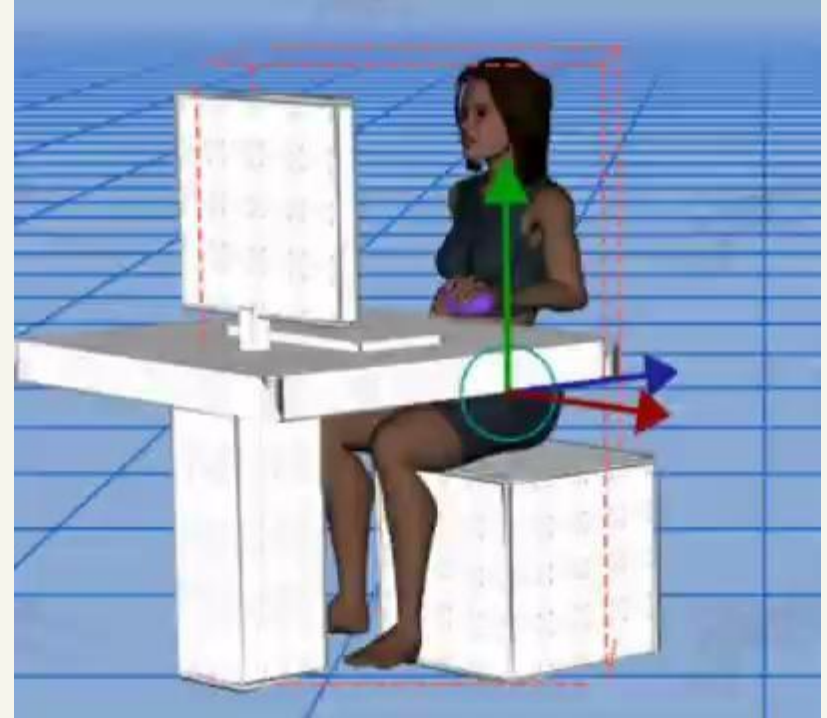
REBA/RULA Findings:

- **RULA Findings:** further investigation, change may be needed
 - Action Level 2: Score 3-4, Investigate Further
- **REBA Findings:** further investigation & implementation soon
 - Score: 4-7, medium risk
- These findings suggest that approaching the problem via designing an ergonomic prototype that provides better support for the wrist is a good idea



SANTOS Video:

- Video shows different possible wrist & mouse postures as analyzed in REBA/RULA assessments



Implementation Plan:

- Most engineering firms have issued tools for their employees
 - Typical cubicle/workspace set up: desk, desktop, basic computer setup etc.
- Companies can provide ergonomic wrist support for workers to use with computer mice



Adherence to Federal Laws Regarding Americans With Disabilities:

- Many engineering firms provide concepts such as spacious workspaces or comfortable desk chairs for their employees so that their health and comfort is not neglected at work. This wrist device could be another form of that. As for ADA, this tool could actually make computer work such as CAD more accessible to workers with disabilities. This tool could be another type of accommodation for those with mobility issues in their joints, blood flow issues, muscle pain, etc.(think of conditions such as thoracic outlet syndrome, Ehlers-Danlos Syndrome, Multiple Sclerosis, etc.)